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Pushbutton switch, operating unit, and amusement machine

Abstract

An amusement machine includes a button deck. The button deck includes a transparent support and a display together serving as a touchscreen, and a pushbutton switch. The pushbutton switch includes an operable portion attached to a first surface of the transparent support, and a detector attached to a second surface of the transparent support and facing the operable portion. The operable portion includes a button. The detector includes a reflective sensor that detects a pressed state of the button through the transparent support.

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Background/Summary

FIELD
(1) The present disclosure relates to a pushbutton switch, an operating unit, and an amusement machine.
BACKGROUND
(2) Patent Literature 1 describes a known pushbutton assembly. The pushbutton assembly described in Patent Literature 1 includes a display and at least one pushbutton switch. The display includes a transparent material on its display surface to support the pushbutton switch. The transparent material is a support for the pushbutton switch.
CITATION LIST
Patent Literature
(3) Patent Literature 1: U.S. Pat. No. 10,431,037
SUMMARY
(4) In the pushbutton assembly described in Patent Literature 1, the pushbutton switch is received in an attachment opening in the transparent material. However, the structure with an opening may lower the strength of the transparent material. When the pushbutton switch is included in an amusement machine for receiving operations, in particular, the transparent material may fracture with cracks extending from the opening.
(5) Accordingly, one or more embodiments is directed to a technique that may reduce fracture of a support in a pushbutton switch.
(6) A switch, a unit, and a machine according to one or more embodiments have the structures described below.
(7) More specifically, a pushbutton switch according to one or more embodiments includes an operable portion attached to a first surface of a support may comprise a plate and may at least partially comprise a transparent area, and a detector attached to a second surface of the support and facing the operable portion with the transparent area being located between the detector and the operable portion. The operable portion includes a press portion, and the detector includes a sensor that detects a pressed state of the press portion through the transparent area.
(8) An operating unit according to one or more embodiments includes a transparent support and a display together serving as a touchscreen, and a pushbutton switch. The pushbutton switch includes an operable portion attached to a surface of the transparent support opposite to the display, and a detector attached to a surface of the transparent support adjacent to the display. The detector faces the operable portion. The operable portion includes a press portion. The detector includes a sensor that detects a pressed state of the press portion through the transparent support.
(9) An amusement machine according to one or more embodiments includes a first display that displays an image for amusement, a transparent support and a second display together serving as a touchscreen, and a pushbutton switch. The pushbutton switch includes an operable portion attached to a surface of the transparent support opposite to the second display, and a detector attached to a surface of the transparent support adjacent to the second display. The detector faces the operable portion. The operable portion includes a press portion. The detector includes a sensor that detects a pressed state of the press portion through the transparent support.
(10) The technique according to one or more embodiments may reduce fracture of the support in the pushbutton switch.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a diagram illustrating an external perspective view of an amusement machine according to one or more embodiments.

(2) FIG. **2** is a diagram illustrating a plan view of a button deck that is an operating unit of an amusement machine, such as is shown in FIG. **1**.

(3) FIG. **3** is a diagram illustrating a cross-sectional view taken along line A-A as viewed in a direction indicated by arrows in FIG. **2**.

(4) FIG. **4** is a diagram illustrating an exploded perspective view of a button deck, such as is shown in FIG. **2**.

(5) FIG. **5** is a diagram illustrating a bottom view of a button deck, such as is shown in FIG. **2**.

(6) FIG. **6** is a diagram illustrating a view example display on a button deck, such as is shown in FIG. **2**.

(7) FIG. **7** is a diagram illustrating an exploded perspective view of an operable portion of a pushbutton switch in a button deck, such as is shown in FIG. **2**.

(8) FIG. **8** is a diagram illustrating a perspective view of a base supporting multiple components in an operable portion, such as is shown in FIG. **7**.

(9) FIG. **9** is a diagram illustrating an exploded perspective view of a detector in a pushbutton switch in a button deck, such as is shown in FIG. **2**.

(10) FIG. **10** is a diagram illustrating a plan view of a mounting board in a detector, such as is shown in FIG. **9**.

(11) FIG. **11** is a block diagram illustrating an amusement machine, such as is shown in FIG. **1**, that includes a control system.

(12) FIG. **12** is a diagram illustrating views of a structure for detecting a pressed state of a button in a pushbutton switch in a button deck, such as is shown in FIG. **2**.

(13) FIG. **13** is a diagram illustrating a cross-sectional view of tabs on a first attachment in a button deck, such as is shown in FIG. **4**.

(14) FIG. **14** is a diagram illustrating cross-sectional views of an engagement portion of a base in an operable portion for engagement with a corresponding tab, such as is shown in FIG. **13**.

(15) FIG. **15** is a diagram illustrating views of a lock assembly for an operable portion, such as is shown in FIG. **7**.

DETAILED DESCRIPTION

(16) One or more embodiments of the present invention (hereafter also referred to as the present embodiment) will now be described with reference to the drawings. In the present embodiment, a pushbutton switch and an operating unit according to an aspect of the present invention are included in an amusement machine. The present invention is not limited to the embodiments described below, and may be variously designed without departing from the spirit and scope of the invention. The pushbutton switch and the operating unit according to an aspect of the present invention can also be used for industrial equipment and consumer equipment, as well as for various amusement machines.

1. Example Use

(17) As shown in FIG. **1**, a pushbutton switch **15** and a button deck **10**, which is an operating unit including the pushbutton switch **15**, may be included in an amusement machine **1**. As shown in FIGS. **3** and **4**, the button deck **10** includes the pushbutton switch **15** and a display input unit **11**. The display input unit **11** includes a transparent support plate **13** and a display **12** and serves as a touchscreen. The pushbutton switch **15** includes, as separate portions, an operable portion **20** and a detector **30**. The operable portion **20** includes a button (a press portion) **21**. The detector **30** detects the pressed state of the button **21**. The operable portion **20** is attached to one surface of the transparent support plate **13**. The detector **30** is attached to the other surface of the transparent support plate **13**. The operable portion **20** and the detector **30** face each other with the transparent

support plate **13** in between. The detector **30** includes sensors for detecting the pressed state of the button **21** through the transparent support plate **13** (with the transparent support plate **13** in between).
(18) The sensors may be, for example, reflective sensors **34** as shown in FIGS. **9** and **12**. As shown in FIGS. **7** and **12**, the operable portion **20** includes reflective flappers **26** that are pressed by a lower end **21***d* of the button **21** and change their orientations when the button **21** is pressed. As shown in views **1000** and **1002** in FIG. **12**, when the button **21** is unpressed, light from a reflective sensor **34** is reflected by a reflective flapper **26** and returns to the reflective sensor **34**. As shown in a view **1001** in FIG. **12**, when the button **21** is pressed, light from the reflective sensor **34** is reflected by the reflective flapper **26** in a direction different from the direction back toward the reflective sensor **34** without returning to the reflective sensor **34**. The reflective sensor **34** receives light reflected by the reflective flapper **26** with changeable intensity, and the change in light intensity allows detection of the pressed state of the button **21**.
(19) The detector **30** may include, for example, range sensors. Using the reflective sensors **34** or the range sensors, electronic components for detecting a pressing operation can be included without being located in the operable portion **20**. However, the electronic components for detecting a pressing operation may be located in the operable portion **20**. Such a structure allows a wider choice of sensors.
(20) The pushbutton switch **15** with the above structure eliminates an opening in the transparent support plate **13** for receiving the pushbutton switch **15**. The transparent support plate **13** can thus reduce fracture due to a decrease in strength when the pushbutton switch **15** is included in the amusement machine **1** for receiving operations.
2. Example Structure
(21) 1. Amusement Machine
(22) The schematic structure of the amusement machine according to the present embodiment will now be described with reference to FIG. **1**. FIG. **1** is an external perspective view of the amusement machine **1** according to the present embodiment.
(23) As shown in FIG. **1**, the amusement machine **1** includes a main display unit (a first display) **2**, a button deck (an operating unit) **10**, and a housing **3** supporting these components.
(24) The main display unit **2** displays images for a game played on the amusement machine **1** and includes, for example, a liquid crystal display (LCD). The main display unit **2** has its display surface facing the front of the amusement machine **1**.
(25) The button deck **10** receives the player's operations in a game on the amusement machine **1**. In the present example, the button deck **10** is located below the main display unit **2** at the front of the amusement machine **1**. The button deck **10** is rectangular and elongated laterally. The button deck **10** includes the display input unit **11** as a touchscreen, and the pushbutton switch **15**.
(26) For the amusement machine **1** providing slot machine gaming involving spinning reels, for example, the main display unit **2** displays multiple reels with multiple types of symbols. The button deck **10** receives the player's instruction for spinning the multiple reels, and receives the number of paylines and the number of bets selected by the player.
(27) The multiple reels start spinning in response to the player's instruction for spinning received by the button deck **10**. The reels automatically stop spinning under the control of a main controller **60** (refer to FIG. **11**, described later). The pattern is determined by the combination of symbols on paylines and by the symbols appearing when the multiple reels are stopped. The player is rewarded based on the determined pattern and the number of bets.
(28) 2. Button Deck
(29) FIG. **2** is a plan view of the button deck **10**. FIG. **3** is a cross-sectional view taken along line A-A as viewed in the direction indicated by arrows in FIG. **2**. FIG. **4** is an exploded perspective view of the button deck **10**. FIG. **5** is a bottom view of the button deck **10**.
(30) As shown in FIG. **2**, the display input unit **11** is rectangular and elongated laterally. The

display input unit **11** includes the pushbutton switch **15** on its display surface. In the present example, the pushbutton switch **15** is circular and at the right end as viewed in plan (refer to FIG. **1**).
(31) As shown in FIG. **3**, the display input unit **11** includes a display (a second display) **12** and a transparent support plate (a support) **13** located on the display surface of the display **12**. In the present example, the transparent support plate **13** is a position input device. The transparent support plate **13** and the display **12** together serve as a touchscreen. The display **12** displays images of keys for receiving inputs to the amusement machine **1**, images for a game, or other images.
(32) The pushbutton switch **15** includes the operable portion **20** and the detector **30**. The operable portion **20** is attached to one surface, specifically a first surface **13***a*, of the transparent support plate **13**. The first surface **13***a* is opposite to the surface facing the display **12**. The operable portion **20** includes the button (a press portion) **21** to receive a pressing operation. The button **21** is transparent.
(33) The detector **30** is attached to the other surface, specifically a second surface **13***b*, of the transparent support plate **13** and faces the operable portion **20**. The second surface **13***b* is adjacent to the display **12**. The detector **30** includes sensors for detecting the pressed state of the button **21** through the transparent support plate **13**.
(34) In the present example, the sensors are reflective sensors **34** (refer to FIG. **9**). However, the sensors may be any sensors other than the reflective sensors **34**, such as range sensors, that can detect the pressed state of the button **21** through the transparent support plate **13**. Electronic components for detecting a pressing operation may be located in the operable portion **20**. Such a structure allows a wider choice of sensors. In the present example, the detector **30** is formed from a transparent material in its area facing the button **21** to allow the display surface of the display **12** to be viewed through the transparent button **21**. The detector **30** may have an opening in its area facing the button **21**.
(35) In the present example, the pushbutton switch **15** is located above the display **12**. The support is the transparent support plate **13** that is entirely transparent except for its outer edge portion. However, the pushbutton switch **15** may not be located above the display **12**. In this case, the support may be transparent simply in its area receiving the operable portion **20** and the detector **30**, or more specifically in its area through which the sensors in the detector **30** detect the pressed state of the button **21**.
(36) In the present example, as shown in FIG. **4**, the operable portion **20** is attached to the first surface **13***a* of the transparent support plate **13** with the first attachment **41** in between. The operable portion **20** is removably attached to the first attachment **41**. The first attachment **41** has an annular shape corresponding to the shape of a bezel **22** (refer to FIG. **7**, described later) in the operable portion **20**. The first attachment **41** is fixed to the first surface **13***a* using a double-sided tape piece **42** in the present example. Attaching the operable portion **20** to the first attachment **41** is described later.
(37) The detector **30** is attached to the second surface **13***b* of the transparent support plate **13** with the second attachment **43** in between. The detector **30** is removably attached to the second attachment **43**. In the present example, the second attachment **43** together with the locking member **45** allows attachment of the detector **30** to the transparent support plate **13**. Each of the second attachment **43** and the locking member **45** has an annular shape and is fitted on the outer periphery of the detector **30**. The second attachment **43** receives the detector **30** on its inner periphery, and the locking member **45** is fitted on the outer periphery of the detector **30**. The locking member **45** is rotated relative to the second attachment **43** and engaged with the second attachment **43**. This attaches the detector **30** to the second surface **13***b*. The second attachment **43** is fixed to the second surface **13***b* using a double-sided tape piece **44** in the present example.
(38) Further, as shown in FIGS. **3** and **5**, the display **12** in the display input unit **11** has its back surface connected to a relay board (a controller) **47** (described later). The relay board **47** is

connected to the detector **30** through a flexible printed circuit (FPC) **37**.
(39) FIG. **6** is a view showing example display on the button deck **10**. The example display in FIG. **6** shows an operation screen for a slot machine game played on the amusement machine **1**. The pushbutton switch **15** serves as a SPIN button that receives the instruction for spinning the multiple reels. The display **12** displays the letters SPIN indicating the SPIN button at the position corresponding to the button **21**. The letters SPIN are viewable through the transparent button **21**. The display input unit **11** displays LINE buttons **17** for selecting the number of paylines and BET buttons **18** for selecting the number of bets.
(40) 3. Pushbutton Switch Structure
(41) 3-1. Operable Portion
(42) The operable portion **20** will now be described with reference to FIGS. **3**, **7**, and **8**. FIG. **7** is an exploded perspective view of the operable portion **20** of the pushbutton switch **15** in the button deck **10**. FIG. **8** is a perspective view of a base **23** supporting multiple components in the operable portion **20**.
(43) As shown in FIG. **7**, the operable portion **20** includes the button **21**, the bezel **22**, the base **23**, multiple springs **24**, multiple tactile-sensation rubber pieces **25**, the locking magnet **27**, and multiple reflective flappers **26**.
(44) The button **21** is supported between the bezel **22** and the base **23** and can receive a pressing operation. For ease of explanation, the operable portion **20** is located horizontally, and the button **21** has the pressing direction being downward and the returning direction being upward.
(45) The button **21** is formed from a transparent material and is circular in the present example (refer to FIG. **2**). The button **21** includes a flange **21***a* on its outer periphery. The button **21** can be pressed with the flange **21***a* being held between the base **23** and the bezel **22** attached to the base **23**. The bezel **22** has an opening **22***a* through which an upper surface **21***b* of the button **21** is exposed.
(46) The base **23** supports, for example, the button **21**, the bezel **22**, the multiple springs **24**, the multiple tactile-sensation rubber pieces **25**, the locking magnet **27**, and the multiple reflective flappers **26**. The base **23** is fastened to the transparent support plate **13** with the first attachment **41** in between.
(47) As shown in FIG. **8**, the base **23** includes an annular component compartment **23***a* in an outer peripheral portion. The component compartment **23***a* accommodates the multiple springs **24**, the multiple tactile-sensation rubber pieces **25**, the locking magnet **27**, and the multiple reflective flappers **26**.
(48) The multiple springs **24** return the pressed button **21** to the position before pressing. The multiple tactile-sensation rubber pieces **25** provide a tactile sensation to the player pressing the button **21**. In the present example, three springs **24** and three tactile-sensation rubber pieces **25** are included.
(49) The multiple springs **24** and the multiple tactile-sensation rubber pieces **25** are in contact with the flange **21***a* on the outer periphery of the button **21** (refer to FIG. **3**). The multiple springs **24** in contact with the flange **21***a* can return the pressed button **21** to its original position. The multiple tactile-sensation rubber pieces **25** in contact with the flange **21***a* can provide a tactile sensation to the player pressing the button **21**.
(50) The reflective flappers **26** are used when the detector **30** detects the pressed state of the button **21**. In the present example, three (multiple) reflective flappers **26** are included. The reflective flappers **26** are supported pivotally on support portions **23***b* of the component compartment **23***a*. Each reflective flapper **26** includes a shaft **26***a* extending in the circumferential direction of the component compartment **23***a*.
(51) Each reflective flapper **26** includes a press piece **26***b* located inward from the shaft **26***a* and a reflective portion **26***c* located outward from the shaft **26***a*. The press piece **26***b* comes in contact with and is pressed by the lower end **21***d* (refer to views **1000** and **1001** in FIG. **12**) of an outer

peripheral portion of the button **21**. The reflective portion **26***c* has its lower surface being a reflective surface. The reflective flapper **26** changes its orientation in accordance with the pressed or unpressed state of the button **21** and thus changes the orientation of the reflective surface (refer to the views **1000** and **1001** in FIG. **12**). The component compartment **23***a* has openings at the bottom to allow passage of light from the reflective sensors **34** in the detector **30** and allow passage of light reflected by the reflective flappers **26** toward the reflective sensors **34**. The first attachment **41** also has such openings.
(52) The locking magnet **27** is rod-like and has the direction of magnetization parallel to the pressing direction of the button **21**. The locking magnet **27** has its lower end fitted in a loose-fitting hole **41***h* (refer to FIG. **15**, described later) in the first attachment **41** to restrict rotation of the operable portion **20** and lock the operable portion **20**. The locking magnet **27** is fitted loosely in a hole **23***c* in the bottom of the component compartment **23***a*. The locking magnet **27** includes a head with a larger diameter than the other portion. The locking magnet **27** is prevented from falling out of the hole **23***c* with its head stuck in the hole **23***c*.
(53) The component compartment **23***a* includes an inner peripheral wall **23***d* defining the component compartment **23***a*. The wall **23***d* guides the button **21** to move vertically when the button **21** is pressed or returns. As shown in FIG. **3**, the button **21** has a groove **21***c* on its lower surface to receive the upper end of the wall **23***d*.
(54) To prepare the operable portion **20**, the button **21** and the bezel **22** are fitted from above in this order onto the component compartment **23***a* accommodating the above various components. The base **23** and the bezel **22** are then fastened together with multiple screws **28**.
(55) 3-2. Detector
(56) The detector **30** will now be described with reference to FIGS. **3**, **9**, and **10**. FIG. **9** is an exploded perspective view of the detector **30** in the pushbutton switch **15** in the button deck **10**. FIG. **10** is a plan view of a mounting board **32**. FIG. **10** also shows an operation lever **36** including an attachment magnet assembly **35**.
(57) As shown in FIG. **9**, the detector **30** includes a case **31**, the mounting board **32**, a lower cover **33**, the multiple reflective sensors **34**, the attachment magnet assembly **35**, the operation lever **36**, and the FPC **37**.
(58) The multiple reflective sensors **34** are mounted on the mounting board **32**. In the present example, three reflective sensors **34** are included to correspond to the three reflective flappers **26** in the operable portion **20**. Each reflective sensor **34** faces the corresponding reflective flapper **26** with the operable portion **20** and the detector **30** being attached to the transparent support plate **13**.
(59) The reflective sensor **34** is an optical sensor including a light emitter **34***a* and a light receiver **34***b*. The light emitter **34***a* emits light toward the reflective flapper **26** through the transparent support plate **13**. The reflective flapper **26** reflects the light that is then received by the light receiver **34***b*. The light receiver **34***b* receives light with intensity changeable in accordance with the orientation of the reflective flapper **26**. The change in light intensity thus allows detection of the pressed state of the button **21**.
(60) As shown in FIG. **10**, the mounting board **32** incorporates multiple light-emitting diodes (LEDs) (light emitters or electronic components) **38** as well as the multiple reflective sensors **34**. The detector **30** can thus emit light when the multiple LEDs **38** is on.
(61) The FPC **37** has one end connected to the mounting board **32** and the other end connected to the relay board **47** (refer to FIG. **3**) attached to the back surface of the display **12**. The FPC **37** electrically connects the mounting board **32** and the relay board **47**.
(62) As shown in FIGS. **9** and **10**, the attachment magnet assembly **35** includes an attraction magnet **35***a* and a repulsion magnet **35***b*. The attraction magnet **35***a* and the repulsion magnet **35***b* both are rod-like and have the direction of magnetization parallel to the pressing direction of the button **21**, but are arranged to have opposite polarities. The attraction magnet **35***a* attracts the locking magnet **27** in the operable portion **20** through the transparent support plate **13**. The

repulsion magnet **35**_b_ repels and lifts the locking magnet **27** in the operable portion **20** through the transparent support plate **13**.
(63) The attraction magnet **35**_a_ and the repulsion magnet **35**_b_ are supported integrally on the operation lever **36**. The operation lever **36** protrudes outward from the outer periphery of the detector **30**. The operation lever **36** is movable back and forth circumferentially along a guide slot **32**_a_ in the mounting board **32** and along a guide slot **33**_a_ in the lower cover **33**. The operation lever **36** is operable to switch the magnet facing the locking magnet **27** in the operable portion **20** between the attraction magnet **35**_a_ and the repulsion magnet **35**_b_. The attachment magnet assembly **35** and the operation lever **36** serve as a switcher. The switcher selectively places the first magnet or the second magnet at the position facing the locking magnet **27** in the operable portion **20** attached to the first attachment **41** (described later).
(64) The case **31** supports, for example, the mounting board **32**, the multiple reflective sensors **34**, the attachment magnet assembly **35**, and the operation lever **36**. The case **31** has openings through which the attachment magnet assembly **35** and the multiple reflective sensors **34** face the locking magnet **27** and the multiple reflective flappers **26** in the operable portion **20**. The lower cover **33** is fitted to the case **31** and covers the bottom of the detector **30**.
(65) To prepare the detector **30**, the lower cover **33** is fitted onto the bottom of the case **31** accommodating, for example, the attachment magnet assembly **35**, the operation lever **36**, and the mounting board **32**. The case **31** and the lower cover **33** are then fastened together with multiple screws **39**.
(66) 4. Control System in Amusement Machine
(67) FIG. **11** is a block diagram of the amusement machine **1**, showing a control system. As shown in FIG. **11**, the amusement machine **1** includes the main controller **60** and the relay board **47** as a controller. The main controller **60** is connected to the main display unit **2** and causes the main display unit **2** to display images for a game to perform the game.
(68) The main controller **60** is also connected to the relay board **47** and receives, using the relay board **47**, an instruction input on the button deck **10**. The main controller **60** also controls the button deck **10** using the relay board **47**. The relay board **47** controls the operation of the button deck **10**, and is connected to the detector **30** through the FPC **37** and also connected to the display input unit **11** as a touchscreen.
(69) As described above, the button deck **10** in the present example includes the pushbutton switch **15** including, as separate portions, the operable portion **20** and the detector **30**. The electronic components are located in the detector **30** alone attached to the second surface **13**_b_ of the transparent support plate **13** adjacent to the inside of the machine. The relay board **47** is thus unconnected to the operable portion **20**.
(70) 5. Detecting Pressing Operation with Pushbutton Switch
(71) FIG. **12** includes views each showing a structure for detecting the pressed state of the button **21** in the pushbutton switch **15** in the button deck **10**. FIG. **12** includes a cross-sectional view **1000** of the main part of the pushbutton switch **15** with the button **21** being unpressed, and includes a cross-sectional view **1001** of the main part of the pushbutton switch **15** with the button **21** being pressed. FIG. **12** also includes a cross-sectional view **1002** of the main part of the pushbutton switch **15** being unpressed. The view **1002** is taken along a line different from the line for the view **1001**.
(72) As shown in the views **1000** and **1002** in FIG. **12**, when the button **21** is unpressed, each reflective flapper **26** in the operable portion **20** is in a first orientation to reflect light from the corresponding reflective sensor **34** in the detector **30** in the direction back toward the reflective sensor **34**. In the first orientation, the reflective surface reflects light from the light emitter **34**_a_ in the direction for entry into the light receiver **34**_b_.
(73) The light receiver **34**_b_ receives, through the transparent support plate **13**, light L emitted from the light emitter **34**_a_ and reflected by the reflective flapper **26** in the first orientation. The light

receiver **34***b* outputs an electrical signal corresponding to the intensity of the received reflected light, and allows detection of the button **21** being unpressed based on the electrical signal.
(74) As shown in the view **1001** in FIG. **12**, when the button **21** is pressed, each reflective flapper **26** in the operable portion **20** is in a second orientation to reflect light from the corresponding reflective sensor **34** in the detector **30** in a direction different from the direction toward the reflective sensor **34**. In the second orientation, the reflective surface reflects light from the light emitter **34***a* in a direction different from the direction for entry into the light receiver **34***b*.
(75) The light receiver **34***b* does not receive light L emitted from the light emitter **34***a* and reflected by the reflective flapper **26** in the second orientation, and thus receives light with reduced intensity. The light receiver **34***b* outputs a lower electrical signal in accordance with the reduced intensity of received light. This change in the electrical signal allows detection of the button **21** being pressed.
(76) 6. Tabs **41***a* on First Attachment **41**
(77) FIG. **13** is a cross-sectional view showing tabs **41***a* on the first attachment **41** in the button deck **10** shown in FIG. **4**. FIG. **14** includes cross-sectional views each showing an engagement portion **20***a* of the base **23** in the operable portion **20** for engagement with the corresponding tab **41***a* shown in FIG. **13**.
(78) As shown in FIG. **13**, the first attachment **41** includes the multiple tabs **41***a* protruding upward. Each tab **41***a* is an L-shaped hook. As shown in FIG. **14**, the base **23** includes the engagement portion **20***a* engageable with the tab **41***a*. The operable portion **20** is rotatable about an axis aligned with the pressing direction from the operable portion **20** toward the detector **30**. This engages the engagement portion **20***a* with the tab **41***a*, thus engaging the base **23** with the first attachment **41**. The base **23** being engaged with the first attachment **41** fastens the operable portion **20** to the first attachment **41**. The base **23** includes multiple engagement portions **20***a* corresponding to the multiple tabs **41***a*.
(79) In the state shown in a view **1003** in FIG. **14**, the tab **41***a* is spaced from the engagement portion **20***a*. When the operable portion **20** is rotated to engage with the first attachment **41**, the engagement portion **20***a* engages with the tab **41***a* as shown in a view **1004** in FIG. **14**. The engagement portion **20***a* being engaged with the tab **41***a* can fasten (attach) the operable portion **20** to the first attachment **41**.
(80) 7. Lock Assembly for Operable Portion **20**
(81) FIG. **15** includes views each showing a lock assembly for the operable portion **20** shown in FIG. **7**. As shown in FIG. **15**, the first attachment **41** has the loose-fitting hole **41***h* for receiving the lower end of the locking magnet **27** with the operable portion **20** being engaged with the first attachment **41**. As shown in a view **1005** in FIG. **15**, when the locking magnet **27** faces the repulsion magnet **35***b* with the transparent support plate **13** in between, the repulsion magnet **35***b* repels and lifts the locking magnet **27**. In this state, the locking magnet **27** is outside the loose-fitting hole **41***h* and allows rotation of the operable portion **20** in the direction for disengagement from the first attachment **41**.
(82) As shown in a view **1006** in FIG. **15**, when the locking magnet **27** faces the attraction magnet **35***a* with the transparent support plate **13** in between, the attraction magnet **35***a* attracts the locking magnet **27**. In this state, the locking magnet **27** has its lower end fitted in the loose-fitting hole **41***h* and locks the operable portion **20** to the first attachment **41**. Operating the above operation lever **36** can switch between the state shown in the view **1005** and the state shown in the view **1006**. This can lock and unlock the operable portion **20** to and from the first attachment **41**.
(83) The operation lever **36** may be constantly urged to the position for locking. With this structure, the attraction magnet **35***a* magnetically attracts the locking magnet **27** and automatically locks the operable portion **20** when the operable portion **20** is attached to the first attachment **41**. In this case, the operation lever **36** may be operated simply for placing the repulsion magnet **35***b* to face the locking magnet **27** to remove the operable portion **20** from the first attachment **41**.
(84) For the pushbutton switch **15** being installed substantially horizontally, the locking magnet **27**

may fall into the loose-fitting hole **41***h* under its weight. Such a structure eliminates the attraction magnet **35***a*.

(85) The operable portion **20** may include multiple locking magnets **27**, and the first attachment **41** may have multiple loose-fitting holes **41***h* corresponding to the multiple locking magnets **27**. In this case, the above switcher selectively places multiple attraction magnets **35***a* or multiple repulsion magnets **35***b* at the positions corresponding to the locking magnets **27**. The multiple attraction magnets **35***a* and the multiple repulsion magnets **35***b* can lock the operable portion **20** to the first attachment **41** more securely.

(86) 8. Effects

(87) As described above, the pushbutton switch **15** with the above structure includes, as fully separate portions, the operable portion **20** and the detector **30** that are located on different surfaces, specifically the front and back surfaces, of the transparent support plate **13**. The transparent support plate **13** thus eliminates an opening for receiving the pushbutton switch and avoids fracture due to a decrease in strength.

(88) The above structure eliminates work for producing an opening in the transparent support plate **13** and facilitates attachment of the pushbutton switch **15**. The above structure also reduces restrictions on the attachment position of the pushbutton switch **15**, increasing flexibility in attachment.

(89) A known pushbutton switch, which includes an operable portion **20** and a detector **30** integral with each other, is to have a different thickness for a transparent support plate **13** with a different thickness (or with a different depth of the attachment opening). In contrast, the pushbutton switch with the above structure is attachable to the transparent support plate **13** with a different thickness that allows the detector **30** to detect the pressed state of the button **21**.

(90) In the above structure, the electronic components for detecting the pressed state of the button **21** can be included without being located in the operable portion **20**. More specifically, the multiple LEDs **38** for lighting the pushbutton switch **15** are located in the detector **30**. The operable portion **20** uses no electricity and eliminates, for example, wiring, thus facilitating maintenance.

(91) In the above structure, the button **21** includes a transparent portion. The detector **30** has an opening or is formed from a transparent material in its area corresponding to the button **21**. This allows the screen of the display **12** to be viewed through the button **21**.

(92) The first attachment **41** and the second attachment **43** facilitate attachment of the operable portion **20** and the detector **30** to the transparent support plate **13**.

(93) Overview

(94) A switch, a unit, and a machine according to one or more aspects of the present invention have the structures described below.

(95) More specifically, a pushbutton switch according to an aspect of the present invention includes an operable portion attached to one surface of a support being a plate and at least partially including a transparent area, and a detector attached to another surface of the support and facing the operable portion with the transparent area being located between the detector and the operable portion. The operable portion includes a press portion, and the detector includes a sensor that detects a pressed state of the press portion through the transparent area.

(96) The above structure includes the sensor for detecting the pressed state of the press portion through the transparent area of the support. Using this sensor, the detector and the operable portion are fully separate from each other and located on different surfaces, specifically the front and back surfaces, of the support. The support thus eliminates an opening for receiving the pushbutton switch and avoids fracture extending from an opening due to a decrease in strength.

(97) The above structure eliminates work for producing an opening in the support and facilitates attachment of the pushbutton switch. The pushbutton switch can be attached at any position in the transparent area of the support. This reduces restrictions on the attachment position and increases flexibility in attachment.

(98) A known pushbutton switch, which includes a detector and an operable portion integral with each other, is to have a different thickness for a support with a different thickness (or with a different depth of the attachment opening). In contrast, the pushbutton switch with the above structure is attachable to the support with a different thickness that allows the detector to detect the pressed state of the press portion.
(99) In the above structure, the electronic component for detecting the pressed state of the press portion can be included without being located in the operable portion.
(100) In the pushbutton switch according to the above aspect, the press portion may include a transparent portion, and the detector may have an opening in an area facing the transparent portion. In another aspect, the press portion may include a transparent portion, and the detector may contain a transparent material in an area facing the transparent portion. In the above structure, the operable portion includes the transparent portion. The detector has an opening or is formed from a transparent material in its area corresponding to the transparent portion. The player can thus view an area farther than the detector through the transparent portion of the operable portion.
(101) The pushbutton switch according to the above aspect may further include an electronic component located in the detector without being located in the operable portion. In the above structure, the operable portion uses no electricity and eliminates, for example, wiring, thus facilitating maintenance.
(102) The pushbutton switch according to one or more embodiments may further include a first attachment fixed to the first surface of the support and removably receiving the operable portion, and a second attachment fixed to the second surface of the support and removably receiving the detector. The above structure facilitates attachment of the operable portion and the detector to the support.
(103) An operating unit according to one or more embodiments includes a transparent support and a display together serving as a touchscreen, and a pushbutton switch. The pushbutton switch includes an operable portion attached to a surface of the transparent support opposite to the display, and a detector attached to a surface of the transparent support adjacent to the display. The detector faces the operable portion. The operable portion includes a press portion. The detector includes a sensor that detects a pressed state of the press portion through the transparent support. The operating unit with the above structure allows its transparent support that supports the pushbutton switch to be resistant to fracture. The operating unit may be included in an amusement machine for receiving operations.
(104) An amusement machine according to one or more embodiments includes a first display that displays an image for amusement, a transparent support and a second display together serving as a touchscreen, and a pushbutton switch. The pushbutton switch includes an operable portion attached to a surface of the transparent support opposite to the second display, and a detector attached to a surface of the transparent support adjacent to the second display. The detector faces the operable portion. The operable portion includes a press portion. The detector includes a sensor that detects a pressed state of the press portion through the transparent support. The amusement machine with the above structure allows its touchscreen that supports the pushbutton switch to be resistant to fracture.
(105) The present invention is not limited to the above embodiments, but may be modified variously within the spirit and scope of the claimed invention. The technical means described in different embodiments may be combined as appropriate in other embodiments within the technical scope of the invention. The technical means described in different embodiments may be combined to produce a new technical feature.

Claims

1. A pushbutton switch, comprising: an operable portion attached to a first surface of a support comprising a plate and at least partially comprising a transparent area, the support extending entirely across an inner face of the operable portion opposite the press portion; a detector attached to a second surface of the support opposite the first surface and facing the operable portion with the transparent area being located between the detector and the operable portion; and a touchscreen display facing the detector and the second surface of the support, wherein the operable portion comprises a press portion on an outer face of the operable portion, the detector is positioned between the second surface of the support and the touchscreen display, and the detector comprises a sensor configured to detect a pressed state of the press portion through the transparent area.
2. The pushbutton switch according to claim 1, wherein the press portion comprises a transparent portion, and the detector comprises an opening in an area facing the transparent portion.
3. The pushbutton switch according to claim 2, further comprising: an electronic component located in the detector without being located in the operable portion.
4. The pushbutton switch according to claim 3, further comprising: a first attachment fixed to the first surface of the support and removably receiving the operable portion; and a second attachment fixed to the second surface of the support and removably receiving the detector.
5. The pushbutton switch according to claim 2, further comprising: a first attachment fixed to the first surface of the support and removably receiving the operable portion; and a second attachment fixed to the second surface of the support and removably receiving the detector.
6. The pushbutton switch according to claim 1, wherein the press portion comprises a transparent portion, and the detector comprises a transparent material in an area facing the transparent portion.
7. The pushbutton switch according to claim 6, further comprising: an electronic component located in the detector without being located in the operable portion.
8. The pushbutton switch according to claim 7, further comprising: a first attachment fixed to the first surface of the support and removably receiving the operable portion; and a second attachment fixed to the second surface of the support and removably receiving the detector.
9. The pushbutton switch according to claim 6, further comprising: a first attachment fixed to the first surface of the support and removably receiving the operable portion; and a second attachment fixed to the second surface of the support and removably receiving the detector.
10. The pushbutton switch according to claim 1, further comprising: an electronic component located in the detector without being located in the operable portion.
11. The pushbutton switch according to claim 10, further comprising: a first attachment fixed to the first surface of the support and removably receiving the operable portion; and a second attachment fixed to the second surface of the support and removably receiving the detector.
12. The pushbutton switch according to claim 1, further comprising: a first attachment fixed to the first surface of the support and removably receiving the operable portion; and a second attachment fixed to the second surface of the support and removably receiving the detector.
13. An operating unit, comprising: a transparent support and a display together serving as a touchscreen; and a pushbutton switch, wherein the pushbutton switch comprises an operable portion attached to a first surface of the transparent support opposite to the display, and a detector attached to a second surface of the transparent support adjacent to the display, the detector facing the operable portion, the operable portion comprises a press portion on an outer face of the operable portion the transparent support extends entirely across an inner face of the operable portion opposite the press portion, the touchscreen facing the detector and the second surface of the transparent support, the detector is positioned between the second surface of the support and the touchscreen display, and the detector comprises a sensor configured to detect a pressed state of the press portion through the transparent support.
14. An amusement machine, comprising: a first display configured to display an image for amusement; a transparent support and a second display together serving as a touchscreen; and a

pushbutton switch, wherein the pushbutton switch comprises an operable portion attached to a first surface of the transparent support opposite to the second display, and a detector attached to a second surface of the transparent support adjacent to the second display, the detector facing the operable portion, the operable portion comprises a press portion on an outer face of the operable portion, the transparent support extends entirely across an inner face of the operable portion opposite the press portion, the touchscreen facing the detector and the second surface of the transparent support, the detector is positioned between the second surface of the support and the touchscreen display, and the detector comprises a sensor configured to detect a pressed state of the press portion through the transparent support.